EXP - 11 (a) ROUTING IN NETWORK UNIT X

a)

AIM:

To simulate static routing configuration using cisco Packet tracer.

PROCEDURE

1. Open cisco packet and create the network topology
   -> Place 3 routers, 3 switches PC's and server in Packet tracer.
   -> Connect the devices as shown in the topology
   -> Configure the required IP addresses for all bites interfaces and end devices

2. Identify the networks.
   -> The networks used are: 10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8 and 50.0.0.0/8
   3 Identify the remote networks that are not directly connected to each router.
   -> Open each router -> CLI
   -> Enter configuration mode
   -> enable.
   -> configure terminal

4. Configure main and backup router.
   -> configure a primary router with a lower administrative Distance (AD)
   -> Configure another router with a higher AD as the backup
   -> If the main route fails, the router automatically uses the backup route.

5. Verify static routing.
   -> check configured static routes using
   -> Test communication using ping and packet

-> Disconnect the main route and verify that packets travel through backup route

6. Delete a static route (destination-network => (subnet-mask> <next-hop>

Result

Thus static routing with main and backup routers was successfully configured using cisco packet tracer. Connectivity between different network was verified using ping and the backup route was successfully used when the main route fiald.

Simulation of RIP using CISCO packet tracer

(b)

AIM:

To simulate and configure RIP (routing information protocol) using CISCO packet tracer and verify communication between different networks.

PROCEDURE:

1- Create the network.

-> place 3 routers (router0, Router 1, Router 2) and 2 PC's (PC0, PC1)

-> connect the routers using serial links and PC's through Fast ethernet interfaces.

-> assign the given IP addresses to all PC's and router Interface.

2. configure Router Interfaces.

-> open router -> CLI and enter: enable

configure terminal interface <interface name>

-> For serial interfaces configure clock rate 64000 and bandwidth 64 where required.

3. Verify connectivity.

open PC -> desktop -> command prompt

Test communication.

verify the PC0 can successfully communicate with PC1 through the router.

4. Configure RIP.

-> enable RIP on each router.

-> advertise all directly connected networks.

-> Repeat the RIP config on router 1 and router 2 with their respective connected networks.

5. Verify Dynamic routing.

=> Use ping to check the status taken by packets

=> Disconnect one routing line and see what occurs

=> RIP automatically selects an alternative route if the current router fails

Result

Thus RIP dynamic routing was successfully configured using Cisco packet tracer communication between different networks was verify using ping and tracert and RIP automatically selected an alternate route when a link failed